

FRANCISTOWN, Botswana

WMO: 680540

Lat: 21.160S

Lon: 27.475E

Elev: 3283

StdP: 13.03

Time Zone: 2.00 (E02)

Period: 96-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.3	38.4	16.0	14.0	50.4	21.9	18.5	53.6	21.1	63.3	18.9	64.5	3.1	20	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	23.7	98.2	66.8	95.6	66.7	93.6	66.6	73.9	84.7	73.0	83.7	72.2	82.9	9.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.5	131.7	77.6	70.2	125.6	76.1	69.7	123.6	75.6	40.0	84.5	39.0	84.3	38.3	82.3	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	18.3	16.5	28.6	101.6	3.9	2.1	25.8	103.2	23.5	104.4	21.2	105.6	18.4	107.2		
(5)			24.8	76.7	3.1	1.7	22.6	78.0	20.8	79.0	19.1	80.0	16.8	81.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.3	77.2	77.0	74.8	69.6	63.5	59.1	58.0	63.6	71.0	76.4	77.9	76.6		
	DBStd	8.48	3.25	3.29	4.00	4.23	4.56	4.69	3.68	5.44	5.26	5.38	5.33	4.39		
	HDD50	3	0	0	0	0	1	2	0	0	0	0	0	0		
	HDD65	603	0	0	1	9	82	183	220	91	12	3	1	1		
	CDD50	7429	843	757	767	589	420	276	247	422	629	817	837	825		
	CDD65	2558	378	337	304	148	37	8	2	48	191	356	388	361		
	CDH74	29350	3321	3185	2716	1670	1148	490	385	1292	2970	4447	4490	3236		
	CDH80	13092	1383	1390	1131	601	325	101	53	444	1489	2364	2353	1458		
(14) Wind	WSAvg	7.4	7.6	7.5	7.2	6.9	5.7	6.0	6.4	7.2	8.4	9.7	8.4	7.8		
(15) Precipitation	PrecAvg	17.8	3.8	3.2	1.7	1.1	0.2	0.1	0.0	0.0	0.3	1.1	2.7	3.3		
	PrecMax	26.4	14.3	13.7	6.3	3.7	3.0	1.4	0.2	0.6	2.4	5.0	6.5	11.5		
	PrecMin	4.6	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.1		
	PrecStd	5.7	3.3	2.9	1.8	0.9	0.6	0.4	0.0	0.1	0.6	1.3	1.5	2.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.4	96.7	96.7	93.3	90.0	86.1	84.5	91.8	98.3	100.8	101.0	96.7		
		MCWB	69.5	68.9	67.0	66.5	61.5	60.0	58.2	61.9	64.7	67.0	68.0	69.3		
	2%	DB	92.9	93.4	93.2	89.7	86.4	82.9	81.2	88.1	95.1	98.2	97.7	93.6		
		MCWB	69.4	68.1	68.6	65.3	61.5	59.7	56.6	59.9	63.0	66.0	67.5	68.7		
	5%	DB	89.9	91.2	89.9	86.8	84.0	79.5	78.6	84.9	92.0	95.3	94.9	91.4		
		MCWB	69.6	69.0	68.7	65.1	60.9	57.8	56.0	59.3	62.1	65.4	66.8	69.3		
	10%	DB	87.7	88.1	86.7	84.0	80.7	76.6	75.4	82.0	89.4	92.6	91.7	88.2		
		MCWB	70.2	68.7	68.4	64.7	59.5	57.3	55.4	58.4	61.6	65.0	67.0	68.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.5	74.5	73.7	72.7	67.5	65.1	61.3	64.8	69.8	71.6	73.0	74.9		
		MCDB	85.9	84.9	85.5	82.2	81.6	77.9	79.9	85.0	87.7	91.2	90.8	86.9		
	2%	WB	74.0	73.3	72.5	70.7	64.8	62.1	59.1	62.7	66.7	69.7	71.5	73.2		
		MCDB	83.7	83.6	84.0	80.4	81.2	76.4	76.9	83.4	86.0	90.3	86.9	85.0		
	5%	WB	73.0	72.3	71.4	68.8	63.0	60.2	57.5	61.1	65.0	68.4	70.4	72.1		
		MCDB	82.2	82.6	82.4	79.2	79.1	74.3	74.0	80.6	85.0	88.6	85.4	83.4		
	10%	WB	72.1	71.2	70.5	67.1	61.2	58.7	56.2	59.6	63.3	67.0	69.3	71.0		
		MCDB	81.1	81.3	80.6	77.8	75.9	72.9	72.1	78.1	83.2	86.0	83.8	81.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.4	20.9	21.1	25.2	33.4	32.7	33.4	33.1	30.7	26.2	23.7	19.5		
		MCDBR	24.8	27.1	27.7	32.9	37.9	39.3	41.6	38.7	37.6	32.9	29.7	26.5		
	5% WB	MCWBR	8.6	9.4	10.0	14.1	17.9	20.8	22.5	18.7	16.2	12.2	10.3	8.6		
		MCWBR	18.3	19.3	20.5	22.7	31.3	32.5	34.7	33.7	31.8	26.9	23.4	19.7		
(40) Clear-Sky Solar Irradiance	taub		0.365	0.362	0.349	0.321	0.289	0.285	0.286	0.337	0.417	0.416	0.385	0.373		
		taud	2.456	2.460	2.492	2.523	2.568	2.547	2.531	2.350	2.153	2.200	2.347	2.429		
	Ebn at Noon		311	307	304	299	298	293	297	290	278	288	302	308		
		Edh at Noon	38	37	35	31	28	27	28	36	48	48	43	39		
(44) All-Sky Solar Radiation	RadAvg		2117	2017	1872	1694	1569	1376	1457	1739	1975	2133	2095	2017		
	RadStd		187	228	185	123	52	94	60	65	97	121	159	179		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page